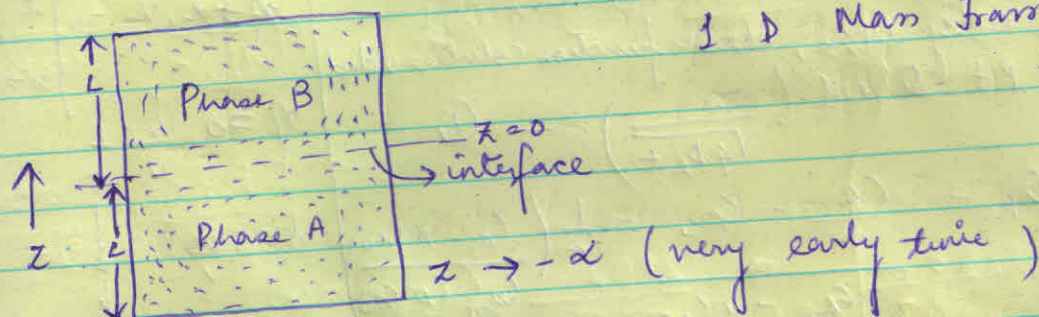


MK - Transient State Mass Transfer



1 D Mass Transfer:

Assumptions -

1. Fick's law is valid (because dilute solution)
2. Interface is at equilibrium $C_B = m C_A$
3. $t \ll \frac{L^2}{D_A}$, $t \ll \frac{L^2}{D_B}$
4. D_A and D_B are of same order.

Initial conditions: $\rightarrow$ (for finite z)

$$C_A = C_{A0} ; C_B = C_{B0}$$

($C_A \rightarrow$ concentration of solute in A phase)

Boundary conditions $\rightarrow$ (for finite t)

$$z \rightarrow \infty, C_B = C_{B0}$$

$$z \rightarrow -\infty, C_A = C_{A0}$$

$$z = 0 \text{ (interface)}, C_B|_{z=0} = m C_A|_{z=0} \quad (m = \text{some constant})$$

Interfacial equilibrium

mass flux continuity at interface,

$$-D_A \frac{\partial C_A}{\partial z} \Big|_{z=0} = -D_B \frac{\partial C_B}{\partial z} \Big|_{z=0}$$

Governing equation:

$$\frac{\partial C_A}{\partial t} = D_A \frac{\partial^2 C_A}{\partial z^2}, \quad -\infty < z < 0$$

$$\frac{\partial C_B}{\partial t} = D_B \frac{\partial^2 C_B}{\partial z^2}, \quad 0 < z < \infty$$

$$\theta_A = \frac{C_A - C_{A0}}{C_{B0} - mC_{A0}} \quad (\text{assuming it will help})$$

For every early time, error function solution valid

$$\theta_A = P + Q \operatorname{erf}\left(\frac{z}{\sqrt{4D_A t}}\right) \quad \text{--- (1)}$$

$$(z=0, \theta_A = \frac{C_B - C_{A0}}{m} - C_{A0} = \frac{1}{m} \left(\frac{C_B - mC_{A0}}{C_{B0} - mC_{A0}} \right)$$

$$\theta_B = \frac{C_B - C_{B0}}{\left(C_{A0} - \frac{C_{B0}}{m} \right)} \quad (\text{non-dimensional conc. of solute in phase B})$$

Analogy with FM

$$\eta \rightarrow \alpha, \frac{v_x}{v} = 0 \quad \left| \begin{array}{l} t \ll \frac{H^2}{v} \\ x \rightarrow \alpha, v_x \end{array} \right.$$

$$\eta \rightarrow 0, \frac{v_x}{v} = 1$$

$$t=0, \frac{v_x}{v} = 0$$

$$\frac{v_x}{v} = 1 - \operatorname{erf}(\eta)$$

Momentum transport in semi-infinite domain

$$\theta_B = R + S \operatorname{erf}\left(\frac{z}{\sqrt{4D_B t}}\right) \quad \text{--- (2)}$$

I.C. $t = 0^+$

$$C_A = C_{A0}$$

$$\theta_A = 0$$

$$C_B = C_{B0}$$

$$\theta_B = 0$$

$$\text{Eqn (1)} : \theta_A = P + Q \Rightarrow 0 = P + Q \Rightarrow P = -Q$$

$$\text{Eqn (2)} : \theta_B = R + S \Rightarrow 0 = R + S \Rightarrow R = -S$$

B.C. $z \rightarrow -\infty, \theta_A = 0$ (same output, $P = Q$) ($C_A = C_{A2}$)
 $z \rightarrow +\infty, \theta_B = 0$ (same output, $R = -S$)

$$z = 0 \quad \theta_A = P + Q \operatorname{erf}(0) \Rightarrow \theta_A = P \Rightarrow \frac{C_{A1} - C_{A0}}{C_{B0} - mC_{A0}} = P$$

$$\theta_B = R + S \operatorname{erf}(0) \Rightarrow \theta_B = R \Rightarrow \frac{C_{B1} - C_{B0}}{C_{A0} - \frac{1}{m}C_{B0}} = R$$

And now, $C_{B1} = mC_{A1}$

$$\text{So, } \frac{mC_{A1} - C_{B0}}{C_{A0} - \frac{C_{B0}}{m}} = R$$

$$\Rightarrow (-m) = \frac{mC_{A1} - C_{B0}}{C_{B0} - mC_{A0}} = R$$

$$x \ll \frac{L^2}{DA} \text{ and } \frac{L^2}{DB} \quad (L \approx \alpha)$$

$$\begin{array}{|c|} \hline C_B \\ \hline \text{Interface} \rightarrow \\ \hline C_A \\ \hline \end{array}$$

$$\text{quasi-steady state (mass flux} = 0) \quad -D_A \frac{\partial C_A}{\partial z} \Big|_{z=0} = -D_B \frac{\partial C_B}{\partial z} \Big|_{z=0}$$

$C_A, C_B \rightarrow$ Solute (common) concentration in phase A & phase B (miscible)

$$Q_A = \frac{C_A - C_{A0}}{C_{B0} - m C_{A0}}, \quad Q_B = \frac{C_B - C_{B0}}{C_{A0} - \frac{C_{B0}}{m}}$$

$$Q_A = P + S \operatorname{erf}\left(\frac{z}{\sqrt{4DA\tau}}\right), \quad Q_B = R + S \operatorname{erf}\left(\frac{z}{\sqrt{4DB\tau}}\right)$$

$P = S \text{ and } R = -S$ from I.C

So, $Q_A = P \left(1 + \operatorname{erf}\left(\frac{z}{\sqrt{4DA\tau}}\right)\right)$ and $Q_B = R \left(1 - \operatorname{erf}\left(\frac{z}{\sqrt{4DB\tau}}\right)\right)$

From B.C., $P = \frac{C_{Ai} - C_{A0}}{C_{B0} - m C_{A0}}$ and $R = \frac{C_{Bi} - C_{B0}}{C_{A0} - \frac{C_{B0}}{m}}$

Now, $C_{Bi} = m C_{Ai}$ (at $z = 0$; interface)

So, $R \left(C_{A0} - \frac{C_{B0}}{m}\right) + C_{B0} = m \left[P \left(C_{B0} - m C_{A0}\right) + C_{A0}\right]$

$\left[\cancel{R C_{A0}} - \cancel{\frac{R C_{B0}}{m}} + C_{B0} = m P C_{B0} - \cancel{m^2 P C_{A0}} + m C_{A0} \right]$

$\Rightarrow -\frac{R}{m} (C_{B0} - m C_{A0}) + C_{B0} = m P (C_{B0} - m C_{A0}) + m C_{A0}$

$\Rightarrow \left(m P + \frac{R}{m}\right) (C_{B0} - m C_{A0}) = (C_{B0} - m C_{A0})$

$\Rightarrow m P + \frac{R}{m} = 1$

$\Rightarrow \boxed{m^2 P + R = m}$

So, $R = m - m^2 P$

Now, $C_A = C_{A0} + (C_{B0} - m C_{A0}) P \left(1 + \exp\left(\frac{z^2}{4DA\tau}\right) \right)$

$$\left. \frac{\partial C_A}{\partial z} \right|_{z=0} = \frac{2P}{\sqrt{\pi}} (C_{B0} - m C_{A0}) - \exp\left(\frac{z^2}{4DA\tau}\right) \frac{1}{\sqrt{4DA\tau}} \Big|_{z=0}$$

$$= \frac{2P (C_{B0} - m C_{A0})}{\sqrt{\pi} \sqrt{4DA\tau}}$$

So, $DA \left. \frac{\partial C_A}{\partial z} \right|_{z=0} = \frac{2PDA}{\sqrt{4\pi DA\tau}} (C_{B0} - m C_{A0})$

Similarly, $DB \left. \frac{\partial C_B}{\partial z} \right|_{z=0} = - \frac{2RDB}{\sqrt{4\pi DB\tau}} \left(C_{A0} - \frac{C_{B0}}{m} \right)$

Thus, $P \sqrt{DA} (C_{B0} - m C_{A0}) = - R \sqrt{DB} \left(C_{A0} - \frac{C_{B0}}{m} \right)$

$$P \sqrt{DA} = \frac{R \sqrt{DB} m}{m}$$

$$P = \frac{R}{m} \sqrt{\frac{DB}{DA}}$$

So, $R = m - m^2 \left[\frac{R}{m} \sqrt{\frac{DB}{DA}} \right] = m - m R \sqrt{\frac{DB}{DA}}$

$$\Rightarrow R \left(1 + m \sqrt{\frac{DB}{DA}} \right) = m \Rightarrow R = \frac{m}{1 + m \sqrt{\frac{DB}{DA}}}$$

So, $P = \frac{1}{1 + m \sqrt{\frac{DB}{DA}}} \left(\sqrt{\frac{DB}{DA}} \right)$

$$= \frac{\sqrt{DB/DA}}{1 + m \left(\sqrt{DB/DA} \right)} = \frac{1}{m + \sqrt{DA/DB}}$$

So, put P and R in ∂A and ∂B .
Find in terms of C_A, C_B .